

Protected Attention–Koopman–T–S Fuzzy References for Normal-Data-Only Fault Detection and Structured Isolation

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Abstract—This paper addresses fault detection and structured isolation for closed-loop nonlinear systems when only normal input–output data are available and no fault direction, fault-input matrix, temporal fault model, or labeled fault sample is known. A central difficulty is that the same measurements are commonly used to form residuals and to update the lifted state, attention context, Takagi–Sugeno (T–S) firing strengths, and nominal reference. An abnormal measurement may consequently move the reference toward the abnormal plant and reduce residual separation. To prevent this feedback, a protected attention–Koopman–T–S fuzzy reference is proposed. A strictly causal attention encoder reads past measured histories, while the nominal branch is initialized from a delay-confirmed normal anchor and subsequently propagated with recorded controller commands, exogenous variables, and predicted outputs only. A start-time-indexed lifted residual unifies free multistep training, short-window monitoring, and long protected-reference monitoring. Its exact nonlinear propagation is derived without differentiating with respect to an unknown fault. The complete fuzzy Koopman Jacobian is decomposed into a weighted local-model term and a firing-strength-variation term, which yields a finite-horizon normal-reference bound governed by rule overlap, local-model dispersion, premise sensitivity, and the realized input trajectory. Based on this propagation, an online dynamic threshold separates the deterministic contributions of anchor uncertainty and normal-model mismatch from a conditionally calibrated stochastic remainder. Full-dimensional multiscale residuals are retained because no fault direction is available for safe projection. Sensor masking and the normal finite-horizon input-response subspace further provide sensor-channel isolation and actuator-compatible/process-side decomposition, with nonunique isolation reported whenever the available information is insufficient.

Index Terms—Attention, dynamic threshold, fault detection and isolation, Koopman operator, normal-data-only learning, Takagi–Sugeno fuzzy systems.

I. INTRODUCTION

CLOSED-LOOP industrial systems frequently operate over broad nonlinear regimes, whereas accurate first-principles models and representative fault data are often unavailable. Learning the normal dynamics and detecting departures from them is therefore attractive for data-driven monitoring. Koopman methods seek lifted coordinates in which nonlinear dynamics admit finite-dimensional linear or scheduling-dependent approximations [9], [10]. Takagi–Sugeno (T–S) fuzzy models preserve local linear structure while describing nonlinear operating-regime dependence through normalized firing strengths [1]–[3].

Recent studies have already combined fuzzy representations with Koopman modeling, including T–S fuzzy Koopman vehicle models and fuzzy deep Koopman multi-model prediction

[13], [14]. Accordingly, this paper does not regard the mere combination of Koopman lifting and T–S interpolation as its contribution. The unresolved monitoring issue considered here is that abnormal measurements may update not only the monitored trajectory but also the learned nominal reference used for comparison.

In many learned observers and predictors, the same measurement is used to construct a residual and to update the lifted state, the premise variables, the fuzzy firing strengths, and the nominal reference. The resulting path can be summarized as

$$\mathbf{y}_k^m \rightarrow \mathcal{H}_k^- \rightarrow \mathbf{z}_k^E \rightarrow \boldsymbol{\rho}_k \rightarrow h_i(\boldsymbol{\rho}_k) \rightarrow F_\Theta. \quad (1)$$

Measurement feedback is beneficial for state tracking, but it can be detrimental to fault detection because the reference should continue to represent normal behavior. The problem is particularly pronounced in an attention–Koopman–T–S architecture: an abnormal measurement can alter the encoded state, the attention context, and the nonlinear interpolation among local Koopman models.

Classical fuzzy fault-detection methods commonly assume explicit fault channels, disturbance descriptions, or prescribed fault-sensitivity objectives and then design observers or filters through Lyapunov and linear-matrix-inequality techniques [4]–[8]. Deep Koopman fault diagnosis has also been studied through learned residuals and parity relations [11], [12]. Those formulations are well suited to known or partly known fault structures. Here, the information boundary is stricter: training uses normal input–output trajectories only, while the fault class, direction, magnitude, temporal profile, and physical input matrix are all unknown. A fault basis, a fault-direction derivative, and a minimum gain along prescribed fault directions are therefore unavailable.

The main contributions are summarized as follows.

- 1) A protected attention–Koopman–T–S fuzzy reference is constructed. The data branch continuously reads strictly past measurements, whereas the nominal branch is initialized only from a delay-confirmed normal anchor and is propagated with the frozen normal model, recorded controller command, measured exogenous variables, and predicted outputs. A start-time-indexed residual provides one mathematical object for free multistep training, short rolling monitoring, and long protected-reference monitoring. Its exact propagation and a sufficient delay-confirmation condition are established without introducing a fault model.

- 2) A fuzzy-structure-dependent finite-horizon analysis is developed. The complete state and input Jacobians are decomposed into weighted local Koopman matrices and a firing-strength-variation term. A dispersion bound makes the effects of rule overlap, local-model disagreement, premise sensitivity, and input variation explicit. The resulting recursion yields a certified normal-reference radius and shows why stable local matrices and one-step prediction accuracy alone do not guarantee a reliable free rollout.
- 3) A normal-data-only detection and structured-isolation scheme is derived. The certified normal-reference recursion generates the deterministic part of an online dynamic threshold, while held-out normal blocks calibrate only the stochastic remainder and the multiscale maximum. A strict residual projection is shown to contain an invisible anomaly direction when no fault direction is known, which motivates full-dimensional residual statistics. Sensor masking and the normal finite-horizon input-response subspace then provide sensor-channel isolation and actuator-compatible/process-side decomposition under explicit identifiability conditions.

The remainder of this paper is organized as follows. Section II gives the Koopman preliminaries and formulates the monitoring problem. Section III presents the protected attention–Koopman–T–S fuzzy model and its normal-data training. Section IV develops the unified residual and protected reference. Section V derives the fuzzy finite-horizon analysis. Section VI presents dynamic detection and structured isolation. Section VII summarizes the offline and online algorithms. Section VIII discusses the role of the fuzzy structure and the information-theoretic limits, and Section IX concludes the paper.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Closed-Loop Plant and Information Boundary

Let $\mathbf{u}_k \in \mathbb{R}^{n_u}$ be the controller command recorded at the controller output, and let $\mathbf{u}_k^a$ denote the action applied to the plant. The nonlinear closed-loop plant is described by

$$\begin{aligned}\mathbf{u}_k^a &= \mathbf{u}_k + \mathbf{f}_k^a, \\ \mathbf{x}_{k+1} &= \mathcal{F}_0(\mathbf{x}_k, \mathbf{u}_k^a, \boldsymbol{\xi}_k) + \mathbf{f}_k^p + \mathbf{w}_k, \\ \mathbf{y}_k^m &= \mathcal{G}_0(\mathbf{x}_k, \mathbf{u}_k^a, \boldsymbol{\xi}_k) + \mathbf{f}_k^s + \mathbf{v}_k,\end{aligned}\quad (2)$$

where $\mathbf{x}_k \in \mathbb{R}^{n_x}$ is the physical state, $\mathbf{y}_k^m \in \mathbb{R}^{n_y}$ is the measured output, and $\boldsymbol{\xi}_k \in \mathbb{R}^{n_\xi}$ is a measured exogenous variable. The vectors $\mathbf{f}_k^a$, $\mathbf{f}_k^p$, and $\mathbf{f}_k^s$ denote unknown actuator-, process-, and sensor-side faults, respectively; $\mathbf{w}_k$ and $\mathbf{v}_k$ collect admissible process and measurement disturbances. No finite-dimensional basis, direction, temporal profile, or differentiable fault-to-output map is assumed.

The nominal model is conditioned on the recorded command $\mathbf{u}_k$. Even if $\mathbf{u}_k$ is generated by a feedback controller that receives abnormal measurements, conditioning both the plant comparison and the normal reference on the same realized command removes the controller map from the conditional difference equation. A trusted measurement of $\mathbf{u}_k^a$, when available, may be used as an additional actuator residual, but it is not required here.

The training set contains normal trajectories $\{\mathbf{u}_k, \mathbf{y}_k^m, \boldsymbol{\xi}_k\}$ only. During monitoring, all model and threshold parameters are frozen. Hence, the objective is not to certify a residual gain for every physical fault. Instead, the design must prevent new abnormal measurements from updating the long nominal reference, bound the fault-free free-rollout error, detect departures from the conditional normal residual set, and restrict isolation claims to structures identifiable from normal data.

B. Controlled Koopman Preliminaries

Consider the normal transition $\mathbf{x}_{k+1} = \mathcal{F}_0(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\xi}_k)$. Let $\mathbb{H}^\infty$ be a Hilbert space of scalar observables $\varphi : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$. For fixed $\mathbf{u}$ and $\boldsymbol{\xi}$, the controlled Koopman operator $\mathcal{U}_{\mathbf{u}, \boldsymbol{\xi}} : \mathbb{H}^\infty \rightarrow \mathbb{H}^\infty$ acts as

$$(\mathcal{U}_{\mathbf{u}, \boldsymbol{\xi}} \varphi)(\mathbf{x}) = \varphi(\mathcal{F}_0(\mathbf{x}, \mathbf{u}, \boldsymbol{\xi})). \quad (3)$$

For a vector-valued lifting $\boldsymbol{\Psi} = [\psi_1, \dots, \psi_{n_z}]^\top$, the exact commutative relation is

$$\boldsymbol{\Psi}(\mathcal{F}_0(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\xi}_k)) = (\mathcal{U}_{\mathbf{u}_k, \boldsymbol{\xi}_k} \boldsymbol{\Psi})(\mathbf{x}_k). \quad (4)$$

A finite-dimensional controlled Koopman model approximates the right-hand side by a linear, bilinear, or scheduling-dependent map. In this paper, the approximation is a T–S interpolation of local lifted linear models. The paper-specific causal history lift and learned firing strengths are introduced in Section III; they are not part of these standard preliminaries. Importantly, the learned lifted state need not contain the physical state explicitly.

C. Monitoring Objectives

The detector is required to satisfy the following conditions.

- 1) New measurements do not update the long nominal reference during a warning or alarm interval.
- 2) The fault-free free-rollout error is propagated over a finite horizon by the complete fuzzy Koopman dynamics.
- 3) The online threshold includes the realized input trajectory, protected fuzzy operating region, reference age, anchor uncertainty, normal-model mismatch, and a separately identified stochastic remainder.
- 4) No residual direction is discarded solely because fault directions are unknown.
- 5) Sensor, actuator-compatible, and process-side labels are reported only under explicit redundancy or subspace-separation conditions.

III. PROTECTED ATTENTION–KOOPMAN–T–S FUZZY MODEL

A. Strictly Causal Attention Encoder

For a history length ℓ_h , define the strictly past history

$$\mathcal{H}_k^- = \{\mathbf{u}_{k-\ell_h:k-1}, \mathbf{y}_{k-\ell_h:k-1}^m, \boldsymbol{\xi}_{k-\ell_h:k}\}. \quad (5)$$

The current measurement $\mathbf{y}_k^m$ is excluded. A temporal attention encoder produces an attention context $\boldsymbol{\chi}_k^E$ and a data-driven lifted state $\mathbf{z}_k^E$:

$$(\mathbf{z}_k^E, \boldsymbol{\chi}_k^E) = \mathcal{E}_\theta(\mathcal{H}_k^-). \quad (6)$$

For completeness, one attention head is represented by

$$\begin{aligned} \mathbf{Q}_m^{\text{att}} &= \mathbf{X}_k \mathbf{W}_m^Q, & \mathbf{S}_m^{\text{att}} &= \mathbf{X}_k \mathbf{W}_m^S, \\ \mathbf{V}_m^{\text{att}} &= \mathbf{X}_k \mathbf{W}_m^V, \\ \mathbf{A}_m^{\text{att}} &= \text{softmax} \left[\frac{\mathbf{Q}_m^{\text{att}} (\mathbf{S}_m^{\text{att}})^\top}{\sqrt{d_m}} + \mathbf{M}^- \right]. \end{aligned} \quad (7)$$

where $\mathbf{M}^-$ is a causal mask and $\mathbf{X}_k$ contains the embedded history tokens. The multihead outputs are aggregated to form χ_k^E and z_k^E . Let $\mathcal{A}_\theta$ denote the context-output submap of $\mathcal{E}_\theta$. Attention is used to capture long temporal dependencies and cross-variable relevance; it is not claimed as a stand-alone contribution.

Because (5) is strictly causal, a sensor deviation occurring at time k has no instantaneous path into z_k^E . It first enters the encoded state through a subsequent history, which is later used for structured isolation.

B. Metric T-S Fuzzy Rules and Local Koopman Models

Let the premise vector be

$$\rho_k = \mathcal{P}_\eta(\text{col}(z_k, \chi_k, \mathbf{u}_k, \xi_k)) \in \mathbb{R}^{n_\rho}. \quad (8)$$

For rule $i \in \{1, \dots, r\}$, define a metric membership score and its normalized firing strength as

$$s_i(\rho) = -\frac{1}{2}(\rho - \mathbf{c}_i)^\top \mathbf{M}_i(\rho - \mathbf{c}_i) + b_i^h, \quad (9)$$

$$h_i(\rho) = \frac{\exp(s_i(\rho))}{\sum_{j=1}^r \exp(s_j(\rho))}, \quad \sum_{i=1}^r h_i = 1, \quad (10)$$

where $\mathbf{M}_i \succ 0$. A bounded-spectrum parameterization can be used to avoid degenerate fuzzy metrics:

$$\mathbf{M}_i = m_{\min} \mathbf{I} + (m_{\max} - m_{\min}) \frac{\mathbf{L}_i \mathbf{L}_i^\top}{\|\mathbf{L}_i\|_F^2 + \epsilon_L}. \quad (11)$$

The local Koopman models and their T-S fuzzy interpolation are

$$\begin{aligned} v_i(z, \mathbf{u}) &= \mathbf{A}_i z + \mathbf{B}_i \mathbf{u} + b_i, \\ \mathcal{F}_\Theta(z, \mathbf{u}, \xi; \chi) &= \sum_{i=1}^r h_i(\rho) v_i(z, \mathbf{u}), \\ \mathcal{G}_\Theta(z, \mathbf{u}) &= \mathbf{C}_z z + \mathbf{D}_z \mathbf{u}, \end{aligned} \quad (12)$$

Unlike a fixed-gate mixture, the firing strengths in (10) vary with the lifted state, attention context, input, and exogenous condition. Their derivatives therefore contribute to the finite-horizon error dynamics.

C. Normal-Data-Only Free-Rollout Training

For a normal anchor s , initialize $\hat{z}_{s|s} = z_s^E$ and set the initial protected history equal to the verified normal history. For $t \geq s$, recursively compute

$$\chi_{t|s}^0 = \mathcal{A}_\theta(\hat{\mathcal{H}}_{t|s}^0), \quad (13)$$

$$\hat{z}_{t+1|s} = \mathcal{F}_\Theta(\hat{z}_{t|s}, \mathbf{u}_t, \xi_t; \chi_{t|s}^0), \quad (14)$$

$$\hat{\mathbf{y}}_{t|s}^0 = \mathcal{G}_\Theta(\hat{z}_{t|s}, \mathbf{u}_t), \quad (15)$$

where $\hat{\mathcal{H}}_{t|s}^0$ is updated with the recorded inputs, exogenous variables, and predicted outputs $\hat{\mathbf{y}}_{t|s}^0$ rather than new measurements. This construction makes the training rollout identical to the protected online rollout.

A representative normal-data objective is

$$\mathcal{L} = \mathcal{L}_{\text{enc}} + \lambda_z \mathcal{L}_z^{\text{ms}} + \lambda_y \mathcal{L}_y^{\text{ms}} + \lambda_m \mathcal{L}_{\text{mask}} + \lambda_f \mathcal{L}_{\text{fuzzy}} + \lambda_J \mathcal{L}_{\text{prop}}, \quad (16)$$

where

$$\mathcal{L}_z^{\text{ms}} = \sum_{j=1}^{N_{\max}} w_j^z \left\| z_{s+j}^E - \hat{z}_{s+j|s} \right\|_{P_z}^2, \quad (17)$$

$$\mathcal{L}_y^{\text{ms}} = \sum_{j=1}^{N_{\max}} w_j^y \left\| \mathbf{y}_{s+j}^m - \hat{\mathbf{y}}_{s+j|s}^0 \right\|_{P_y}^2. \quad (18)$$

The rollout is free-running; teacher forcing is not applied at intermediate prediction steps. The fuzzy regularizer discourages rule collapse and ill-conditioned metrics, while $\mathcal{L}_{\text{prop}}$ penalizes excessive finite-horizon Jacobian products.

IV. PROTECTED FUZZY REFERENCE AND UNIFIED RESIDUAL

A. Start-Time-Indexed Residual

For any anchor $s \leq k$, define

$$e_{k|s}^z = z_k^E - \hat{z}_{k|s}. \quad (19)$$

This single definition serves three purposes:

- 1) normal multistep training uses random (s, k) pairs;
- 2) short-window monitoring uses $s = k - L_s$;
- 3) long-horizon monitoring uses the latest confirmed safe anchor s_k^0 .

Accordingly,

$$\mathbf{r}_k^S = e_{k|k-L_s}^z, \quad \mathbf{r}_k^L = e_{k|s_k^0}^z, \quad \mathbf{z}_k^0 = \hat{\mathbf{z}}_{k|s_k^0}. \quad (20)$$

Training and monitoring therefore use the same residual and differ only in the anchor-selection rule.

B. Delayed Confirmation and Recovery

A newly encoded state is not accepted immediately as a safe anchor. It first enters a candidate queue and is confirmed only after D_c subsequent samples remain below the warning thresholds. When a warning is issued, all unconfirmed candidates are discarded and the long reference continues from the latest confirmed anchor.

Proposition 1 (Blocking pre-alarm contamination). *Suppose an abnormal interval begins at time τ and the warning/alarm logic reacts within at most D_a samples for the abnormality under consideration. If $D_c \geq D_a$, then a candidate whose encoder history first contains the abnormal measurement at τ cannot be confirmed before the alarm.*

Proof. By strict causality, the earliest affected encoded state is $z_{\tau+1}^E$. Its earliest confirmation time is $\tau+1+D_c > \tau+D_a$. $\square$

If the abnormal interval ends at τ_e , the encoder history is free of abnormal measurements only after ℓ_h additional samples. Anchor acceptance is therefore re-enabled only after $\ell_h + D_c$ normal samples.

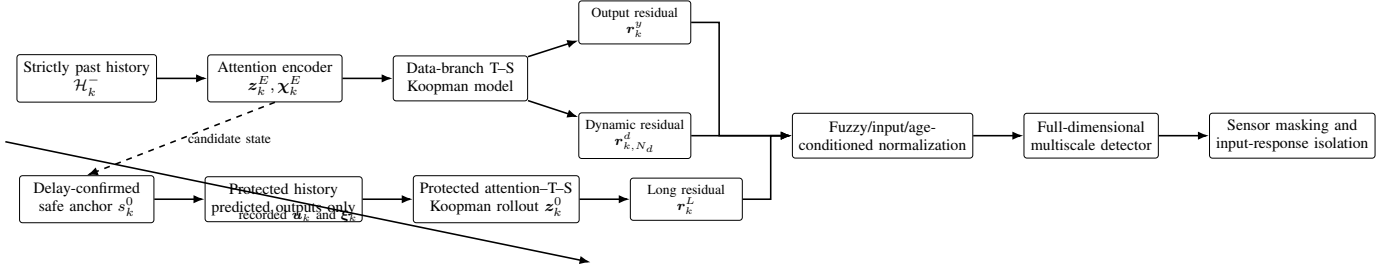


Fig. 1. Protected attention-Koopman-T-S fuzzy monitoring architecture. Measurements continuously enter the data branch and residual computation. The long nominal reference is propagated from a delay-confirmed safe anchor using predicted outputs, the recorded controller command, and the frozen fuzzy Koopman model; hence, new measurements do not update its attention context or firing strengths during monitoring.

C. Exact Residual Propagation Without a Fault Model

Define the computable one-step innovation of the data branch as

$$\nu_k^E = z_{k+1}^E - \mathcal{F}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E). \quad (21)$$

The difference between the data context and the protected context is represented by

$$\delta_{k|s}^X = \mathcal{F}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E) - \mathcal{F}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^0). \quad (22)$$

Theorem 1 (Exact propagation of the unified residual). *Assume that $\mathcal{F}_\Theta$ is differentiable on the segment joining $\hat{z}_{k|s}$ and z_k^E . Then*

$$e_{k+1|s}^z = \bar{A}_{k|s} e_{k|s}^z + \nu_k^E + \delta_{k|s}^X, \quad (23)$$

where

$$\bar{A}_{k|s} = \int_0^1 J_z \mathcal{F}_\Theta(\hat{z}_{k|s} + \tau e_{k|s}^z, u_k, \xi_k; \chi_{k|s}^0) d\tau. \quad (24)$$

Consequently,

$$e_{k|s}^z = \bar{\Phi}(k, s) e_{s|s}^z + \sum_{j=s}^{k-1} \bar{\Phi}(k, j+1) (\nu_j^E + \delta_{j|s}^X), \quad (25)$$

with $\bar{\Phi}(k, j) = \bar{A}_{k-1|s} \cdots \bar{A}_{j|s}$.

Proof. Insert (21) into $z_{k+1}^E - \hat{z}_{k+1|s}$, add and subtract $\mathcal{F}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^0)$, and apply the integral mean-value formula to the state difference. Repeated substitution gives (25). $\square$

The theorem does not assume a fault direction or differentiate with respect to a fault variable. It shows that the multistep residual is the finite-horizon accumulation of directly computable one-step innovations and context mismatch through the normal fuzzy Koopman dynamics.

Remark 1 (Attention memory in the analysis). *The Jacobian in (24) is the partial state Jacobian at the protected attention context. The non-Markovian effect of the finite history buffer is not silently absorbed into that Jacobian; it appears explicitly through $\delta_{k|s}^X$ and is bounded as part of the normal forcing term in Assumption 1. An equivalent augmented-state proof can be obtained by appending the protected history buffer to the lifted state.*

V. FUZZY-STRUCTURE-DEPENDENT FINITE-HORIZON ANALYSIS

A. Complete Jacobian of the Fuzzy Koopman Map

Let $v_i = v_i(z, u)$, $h_i = h_i(\rho)$, and define

$$\bar{v} = \sum_{i=1}^r h_i v_i, \quad (26)$$

$$g_i(\rho) = \nabla_\rho s_i(\rho) = -M_i(\rho - c_i), \quad (27)$$

$$\bar{g} = \sum_{i=1}^r h_i g_i. \quad (28)$$

Theorem 2 (Complete Jacobians of the T-S fuzzy Koopman map). *The state and input Jacobians of (12) are*

$$J_z \mathcal{F}_\Theta = A_h + C_\rho J_{\rho,z}, \quad (29)$$

$$J_u \mathcal{F}_\Theta = B_h + C_\rho J_{\rho,u}, \quad (30)$$

where

$$A_h = \sum_{i=1}^r h_i A_i, \quad B_h = \sum_{i=1}^r h_i B_i, \quad (31)$$

$$C_\rho = \sum_{i=1}^r h_i (v_i - \bar{v})(g_i - \bar{g})^\top. \quad (32)$$

Define the local-model and premise-gradient dispersion matrices

$$V = \sum_{i=1}^r h_i (v_i - \bar{v})(v_i - \bar{v})^\top, \quad (33)$$

$$G = \sum_{i=1}^r h_i (g_i - \bar{g})(g_i - \bar{g})^\top. \quad (34)$$

Then

$$\|C_\rho J_{\rho,z}\|_2 \leq \sqrt{\lambda_{\max}(V)} \sqrt{\lambda_{\max}(J_{\rho,z}^\top G J_{\rho,z})}. \quad (35)$$

The corresponding input bound follows by replacing $J_{\rho,z}$ with $J_{\rho,u}$.

Proof. The normalized weights satisfy $\nabla_\rho h_i = h_i(g_i - \bar{g})$. Substitution into the derivative of $\sum_i h_i v_i$ gives (29)–(32). The bound follows from the weighted Cauchy-Schwarz inequality; details are given in Appendix A. $\square$

Remark 2 (Fuzzy interpretation). *The second term in (29) is nonzero only when three ingredients coexist: several rules have*

nonnegligible firing strengths, their local Koopman outputs disagree, and the considered state or input direction changes their relative firing strengths. Thus, stable local matrices $\mathbf{A}_i$ do not by themselves guarantee a reliable free rollout. The fuzzy interpolation mechanism is part of the error dynamics rather than a static model-selection layer.

B. Finite-Horizon Normal Error Bound

Define the computable local amplification bound

$$L_k = \|\mathbf{A}_{h,k}\|_2 + \sqrt{\lambda_{\max}(\mathbf{V}_k)} \sqrt{\lambda_{\max}(\mathbf{J}_{\rho,z,k}^\top \mathbf{G}_k \mathbf{J}_{\rho,z,k})}. \quad (36)$$

Assumption 1 (Normal operating domain). *The frozen model is continuously differentiable on a compact normal domain $\mathcal{D}$. Every segment in (24) remains in $\mathcal{D}$, and the normal forcing term satisfies $\|\nu_k^{E,0} + \delta_{k|s}^{X,0}\|_2 \leq \epsilon_k$.*

Theorem 3 (Finite-horizon normal free-rollout bound). *If $\|\mathbf{J}_z \mathcal{F}_\Theta\|_2 \leq L_k$ on the relevant segments and $\mathbf{e}_{s|s}^z = \mathbf{0}$, then*

$$\|\mathbf{e}_{s+\ell|s}^{z,0}\|_2 \leq \sum_{j=0}^{\ell-1} \left(\prod_{p=j+1}^{\ell-1} L_{s+p} \right) \epsilon_{s+j}. \quad (37)$$

If $L_k \leq L$ and $\epsilon_k \leq \epsilon$, the right-hand side equals $\epsilon(1 - L^\ell)/(1 - L)$ for $L < 1$, $\ell\epsilon$ for $L = 1$, and $\epsilon(L^\ell - 1)/(L - 1)$ for $L > 1$.

Proof. Apply the norm to (23) under normal operation and recursively substitute the resulting inequality. $\square$

For a tolerated normal latent-residual radius b_c in fuzzy operating cell c , define the reliable rollout horizon

$$N_c^* = \max \left\{ N : \sum_{j=0}^{N-1} \left(\prod_{p=j+1}^{N-1} L_{s+p} \right) \epsilon_{s+j} \leq b_c \right\}. \quad (38)$$

When the reference age exceeds N_c^* , the detector must use an age-dependent threshold, switch to a more conservative normal model, or refrain from claiming a bounded nominal reference.

VI. DYNAMIC DETECTION AND STRUCTURED ISOLATION

A. Complementary Residuals

Three residual blocks are retained:

$$\mathbf{r}_k^y = \mathbf{y}_k^m - \mathcal{G}_\Theta(\mathbf{z}_k^E, \mathbf{u}_k), \quad (39)$$

$$\mathbf{r}_{k,N_d}^d = \text{col}(\nu_{k-N_d+1}^E, \dots, \nu_k^E), \quad (40)$$

$$\mathbf{r}_k^L = \mathbf{e}_{k|s^0}^z. \quad (41)$$

The output residual reacts directly to a current sensor deviation because $\mathbf{z}_k^E$ is strictly causal. The dynamic block captures recent departures from the learned normal dynamics. The long residual accumulates persistent deviations from the latest safe reference. Their joint vector is

$$\mathbf{R}_k = \text{col}(\mathbf{r}_k^y, \mathbf{r}_{k,N_d}^d, \mathbf{r}_k^L). \quad (42)$$

TABLE I
TYPICAL RESIDUAL PROPAGATION SIGNATURES

Type	$\mathbf{r}^y$	$\mathbf{r}^d$	$\mathbf{r}^L$
Sensor	Immediate direct response	Delayed through the encoder history	Delayed and accumulated
Actuator	Depends on plant relative degree	Early input-compatible response	Accumulated
Process	Depends on output path	General dynamic direction	Accumulated

B. Why the Detection Statistic Remains Full Dimensional

Proposition 2 (Invisible direction under strict projection). *Let the normalized normal residual satisfy $\tilde{\mathbf{R}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_m)$ and let an anomaly produce $\tilde{\mathbf{R}} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{I}_m)$ with unspecified $\boldsymbol{\mu}$. For every rank- q linear projection $\mathbf{Q} \in \mathbb{R}^{q \times m}$ with $q < m$, there exists $\boldsymbol{\mu} \neq \mathbf{0}$ such that $\mathbf{Q}\boldsymbol{\mu} = \mathbf{0}$. The projected normal and abnormal distributions are then identical. By contrast, the full statistic $\|\tilde{\mathbf{R}}\|_2^2$ is monotone in the generalized likelihood ratio after maximizing over $\boldsymbol{\mu}$.*

Proof. Because $q < m$, $\ker(\mathbf{Q})$ contains a nonzero vector. Selecting $\boldsymbol{\mu} \in \ker(\mathbf{Q})$ yields identical projected means and covariances. Maximization of the Gaussian likelihood over $\boldsymbol{\mu}$ gives $\hat{\boldsymbol{\mu}} = \tilde{\mathbf{R}}$ and a likelihood ratio monotone in $\|\tilde{\mathbf{R}}\|_2^2$. $\square$

Thus, fault-basis-driven dimensionality reduction is inconsistent with the present no-fault-prior setting. Projection is used only after an alarm for mechanism interpretation, where the projection subspace is supplied by the normal input-response structure.

C. Residual-Dynamics-Based Online Threshold

The term *dynamic threshold* is used here for an online threshold generated from the fault-free residual dynamics, rather than for a moving empirical quantile. The conditional normalizer and held-out quantiles introduced below calibrate only the stochastic remainder.

Let the protected operating descriptor be

$$\zeta_k = \text{col}(\mathbf{h}_k^0, \bar{\mathbf{h}}_{k,N}^0, \mathbf{u}_{k-N:k}, \Delta \mathbf{u}_{k-N:k}, \boldsymbol{\xi}_{k-N:k}, a_k), \quad (43)$$

where $a_k = k - s_k^0$ is the reference age. Normal data are used to estimate a conditional mean $\boldsymbol{\mu}_R(\zeta_k)$ and a positive-definite covariance $\boldsymbol{\Sigma}_R(\zeta_k)$. For a fuzzy mixture with local weights ω_c , means $\boldsymbol{\mu}_c$, and covariances $\boldsymbol{\Sigma}_c$, the covariance is

$$\begin{aligned} \boldsymbol{\mu}_R &= \sum_c \omega_c \boldsymbol{\mu}_c, \\ \boldsymbol{\Sigma}_R &= \sum_c \omega_c [\boldsymbol{\Sigma}_c + (\boldsymbol{\mu}_c - \boldsymbol{\mu}_R)(\boldsymbol{\mu}_c - \boldsymbol{\mu}_R)^\top] + \epsilon_\Sigma \mathbf{I}. \end{aligned} \quad (44)$$

The between-cell term in (44) is required for the covariance of the fuzzy mixture. Define $\mathbf{W}_k = \boldsymbol{\Sigma}_R(\zeta_k)^{-1/2}$.

Assumption 2 (Fault-free residual decomposition). *On every certified fuzzy operating cell, the centered fault-free joint residual admits*

$$\mathbf{R}_k^0 - \boldsymbol{\mu}_R(\zeta_k) = \mathbf{d}_k^R + \mathbf{n}_k^R, \quad (45)$$

where $\mathbf{d}_k^R$ contains bounded anchor and normal-model mismatch terms and $\mathbf{n}_k^R$ is the stochastic remainder. The following deterministic envelopes are available from the frozen normal model and normal validation data:

$$\begin{aligned} \|\mathbf{e}_{s|s}^{z,0}\|_2 &\leq \bar{\epsilon}_s, & \|\mathbf{r}_k^{y,0} - \mathbf{n}_k^y\|_2 &\leq \bar{\epsilon}_k^y, \\ \|\boldsymbol{\nu}_k^{E,0} - \mathbf{n}_k^d\|_2 &\leq \bar{\epsilon}_k^d, & \|\boldsymbol{\nu}_k^{E,0} + \boldsymbol{\delta}_{k|s}^{X,0} - \mathbf{n}_k^z\|_2 &\leq \bar{\epsilon}_k^z. \end{aligned} \quad (46)$$

The bounds $\bar{\epsilon}_k^y$, $\bar{\epsilon}_k^d$, and $\bar{\epsilon}_k^z$ may depend on ζ_k and therefore on the realized input trajectory and protected fuzzy firing strengths.

Starting from $\beta_{s|s}^z = \bar{\epsilon}_s$, propagate the deterministic long-reference radius by

$$\beta_{k+1|s}^z = L_k \beta_{k|s}^z + \bar{\epsilon}_k^z. \quad (47)$$

For the residual blocks in (42), define

$$\begin{aligned} \beta_k^y &= \bar{\epsilon}_k^y, \\ \beta_{k,N_d}^d &= \left(\sum_{j=k-N_d+1}^k (\bar{\epsilon}_j^d)^2 \right)^{1/2}, \\ \beta_k^L &= \beta_{k|s}^z, & \beta_k^R &= [(\beta_k^y)^2 + (\beta_{k,N_d}^d)^2 + (\beta_k^L)^2]^{1/2}. \end{aligned} \quad (48)$$

The normalized deterministic contribution is bounded by

$$\gamma_k^{\text{det}} = \|\mathbf{W}_k\|_2 \beta_k^R. \quad (49)$$

Let $q_{c,a,1-\alpha}$ be a finite-sample corrected conditional quantile of $\|\mathbf{W}_k \mathbf{n}_k^R\|_2$ obtained from nonoverlapping held-out normal blocks after the model, envelopes, fuzzy cells, and normalizer have been frozen. The online instantaneous dynamic threshold is

$$J_k^{\text{dyn}} = \gamma_k^{\text{det}} + q_{c_k,a_k,1-\alpha}. \quad (50)$$

An alarm is issued when

$$\|\mathbf{W}_k[\mathbf{R}_k - \boldsymbol{\mu}_R(\zeta_k)]\|_2 > J_k^{\text{dyn}}. \quad (51)$$

Theorem 4 (Pointwise fault-free threshold guarantee). *Suppose Assumption 2 holds and, conditionally on cell c_k and age group a_k , the held-out calibration remainders and the next fault-free remainder are exchangeable. Then*

$$\Pr\left\{\|\mathbf{W}_k[\mathbf{R}_k^0 - \boldsymbol{\mu}_R(\zeta_k)]\|_2 \leq J_k^{\text{dyn}}\right\} \geq 1 - \alpha. \quad (52)$$

Proof. The triangle inequality, (49), and the corrected calibration quantile give $\|\mathbf{W}_k(\mathbf{d}_k^R + \mathbf{n}_k^R)\|_2 \leq \gamma_k^{\text{det}} + \|\mathbf{W}_k \mathbf{n}_k^R\|_2$. The claimed probability follows from the conditional finite-sample quantile property. $\square$

For multiscale monitoring, define the stacked residual $\mathcal{R}_{k,N} = \text{col}(\mathbf{R}_{k-N+1}, \dots, \mathbf{R}_k)$ and use a full stacked normalizer $\mathcal{W}_{k,N}$ that retains cross-time covariance. The deterministic stacked radius is

$$\Gamma_{k,N}^{\text{det}} = \|\mathcal{W}_{k,N}\|_2 \left(\sum_{j=k-N+1}^k (\beta_j^R)^2 \right)^{1/2}. \quad (53)$$

The recursion (47) is evaluated at every time in the stack; hence, cross-time propagation of anchor uncertainty and model mismatch is not discarded. Define the multiscale excess score

$$S_k^{\text{max}} = \max_{N \in \mathcal{N}} [\|\mathcal{W}_{k,N}(\mathcal{R}_{k,N} - \boldsymbol{\mu}_{k,N})\|_2 - \Gamma_{k,N}^{\text{det}}]. \quad (54)$$

A single held-out quantile of S_k^{max} controls the familywise window-wise false-alarm probability under the corresponding block-exchangeability assumption. This quantile is an auxiliary stochastic calibration; the online time variation in the threshold is generated by (47)–(53).

D. Sensor-Channel Isolation

A shared mask-conditioned encoder is trained on normal data:

$$(\mathbf{z}_k^{E,M}, \boldsymbol{\chi}_k^{E,M}) = \mathcal{E}_\vartheta(\mathcal{H}_k^-, \mathbf{M}). \quad (55)$$

For sensor j , let $\mathbf{M}_{-j}$ mask its history and define

$$r_{j,k}^s = y_{j,k}^m - G_{\Theta,j}(\mathbf{z}_k^{E,-j}, \mathbf{u}_k). \quad (56)$$

A dominant early increase in $r_{j,k}^s$ supports a sensor- j decision, provided the remaining channels and inputs retain enough information to predict that sensor. When this redundancy condition is not met, the method reports an unresolved sensor-channel anomaly rather than forcing a label.

E. Actuator-Compatible and Process-Side Components

The complete normal input Jacobian follows from (30):

$$\mathbf{B}_k^u = \mathbf{J}_u \mathcal{F}_\Theta = \mathbf{B}_{h,k} + \mathbf{C}_{\rho,k} \mathbf{J}_{\rho,u,k}. \quad (57)$$

For a horizon N_a , define the block input-response matrix $\mathcal{B}_{k,N_a}$ with blocks

$$[\mathcal{B}_{k,N_a}]_{\ell j} = \begin{cases} \Phi_A(t_\ell, s_j + 1) \mathbf{B}_{s_j}^u, & j < \ell, \\ \mathbf{0}, & j \geq \ell, \end{cases} \quad (58)$$

where $\mathbf{A}_t^z = \mathbf{J}_z \mathcal{F}_\Theta$ and $\Phi_A(t, s) = \mathbf{A}_{t-1}^z \cdots \mathbf{A}_s^z$. Let $\tilde{\mathcal{B}}_{k,N_a}$ and $\tilde{\mathbf{r}}_{k,N_a}^d$ denote their conditionally whitened versions. The projector onto the normal input-response subspace is

$$\mathbf{P}_{u,k} = \tilde{\mathcal{B}}_{k,N_a} (\tilde{\mathcal{B}}_{k,N_a}^\top \tilde{\mathcal{B}}_{k,N_a})^\dagger \tilde{\mathcal{B}}_{k,N_a}^\top. \quad (59)$$

Define

$$T_k^a = \|\mathbf{P}_{u,k} \tilde{\mathbf{r}}_{k,N_a}^d\|_2^2, \quad (60)$$

$$T_k^p = \|(\mathbf{I} - \mathbf{P}_{u,k}) \tilde{\mathbf{r}}_{k,N_a}^d\|_2^2, \quad (61)$$

$$\hat{\mathbf{f}}_{k,N_a}^{a,\text{eq}} = (\tilde{\mathcal{B}}_{k,N_a}^\top \tilde{\mathcal{B}}_{k,N_a})^\dagger \tilde{\mathcal{B}}_{k,N_a}^\top \tilde{\mathbf{r}}_{k,N_a}^d. \quad (62)$$

Here, T_k^a measures the dynamic residual component compatible with an input deviation, whereas T_k^p measures the component not explainable by the normal input channel. The estimate in (62) is therefore an *equivalent input deviation*; it is not automatically a physical actuator-fault estimate.

Theorem 5 (Actuator/process separation boundary). *Suppose the whitened dynamic residual satisfies*

$$\tilde{\mathbf{r}}^d = \tilde{\mathcal{B}} \mathbf{f}^a + \mathbf{f}^p + \boldsymbol{\epsilon}, \quad (63)$$

where $\mathbf{f}^p \perp \text{Range}(\tilde{\mathcal{B}})$. If $\tilde{\mathcal{B}}$ has full column rank, the least-squares estimate is unique and the parallel and orthogonal projections recover the actuator-compatible and process-side components up to $\boldsymbol{\epsilon}$. If $\mathbf{f}^p$ contains a component in $\text{Range}(\tilde{\mathcal{B}})$, that component cannot be distinguished from an actuator deviation using the available input-output information.

Algorithm 1 Normal-Data-Only Offline Design

Require: Normal trajectories $\{u_k, y_k^m, \xi_k\}$

- 1: Train the strictly causal attention encoder and obtain z_k^E and χ_k^E .
 - 2: Jointly train the metric T-S fuzzy rules, local Koopman models, and decoder.
 - 3: Optimize free-running multistep latent and output losses using the protected-history rollout.
 - 4: Train the shared mask-conditioned encoder on normal sensor masks.
 - 5: Generate multi-anchor residuals and estimate fuzzy/input/age-conditioned moments.
 - 6: Evaluate the complete Jacobians and reliable rollout horizons.
 - 7: Calibrate the stochastic remainders, the multiscale maximum, and the structured-isolation scores on held-out normal blocks.
 - 8: Freeze all learned and statistical parameters.
-

Algorithm 2 Protected Fuzzy-Reference Online Monitoring

Require: Frozen model, latest safe anchor, candidate queue, dynamic-threshold parameters and stochastic calibrations

- 1: **for** each sample k **do**
 - 2: Compute z_k^E and χ_k^E from strictly past data.
 - 3: Propagate the short rollouts and the protected long reference using u_k and ξ_k .
 - 4: Compute $r_k^y, r_{k,N_d}^d, r_k^L$, and ζ_k .
 - 5: Evaluate the instantaneous dynamic threshold and the multiscale excess score.
 - 6: **if** the warning threshold is exceeded **then**
 - 7: Freeze anchor updates and discard unconfirmed candidates.
 - 8: **end if**
 - 9: **if** the alarm threshold is exceeded **then**
 - 10: Evaluate the masked sensor residuals.
 - 11: Evaluate T_k^a, T_k^p , and $\hat{f}_{k,N_a}^{a,eq}$.
 - 12: Report sensor-channel, actuator-compatible, process-side, or nonunique isolation.
 - 13: **else**
 - 14: Add z_k^E to the delayed-confirmation queue.
 - 15: **end if**
 - 16: After recovery, wait $\ell_h + D_c$ normal samples before accepting new anchors.
 - 17: **end for**
-

For actuator j , let $\mathcal{B}_j$ be its finite-horizon response matrix and let $\mathcal{B}_{-j}$ collect all remaining actuators. The condition

$$\text{rank}[\mathcal{B}_j, \mathcal{B}_{-j}] > \text{rank}(\mathcal{B}_{-j}) \quad (64)$$

ensures that actuator j has at least one response direction not reproducible by the other actuator channels.

VII. OFFLINE AND ONLINE DESIGN

The online workload consists of attention inference, blending r local Koopman models, maintaining residual buffers, evaluating the conditional normalizer, and recursively updating

a finite-horizon input-response matrix. Complete Jacobians may be evaluated less frequently or through Jacobian–vector products. Isolation is activated only after an alarm. No online training, clustering, semidefinite programming, or fault-sample classifier is required.

VIII. DISCUSSION

A. Fuzzy Contribution

The contribution is not the juxtaposition of a fuzzy rule base and a Koopman model. The fuzzy structure is essential in four specific places. First, the protected branch prevents abnormal measurements from updating the nominal firing strengths. Second, the complete Jacobian in (29) quantifies the additional error amplification caused by rule overlap, local-model dispersion, and premise sensitivity. Third, the complete fuzzy dynamics generate the input- and region-dependent deterministic threshold component, while the conditional stochastic remainder is calibrated within protected fuzzy operating regions. Fourth, the input-response geometry includes the change of the firing strengths with the input through $C_\rho J_{\rho,u}$. Treating the weights as fixed gates removes all four properties.

B. Detectability and Identifiability

Without a fault-effect prior, no universal lower bound can map a physical fault magnitude to a residual magnitude. Detection is guaranteed only for deviations that move the conditionally normalized joint residual outside the calibrated normal region. Internal abnormalities that are invisible at the measured input–output level, deviations that remain on the normal behavior set, and collisions in the learned representation cannot be detected by any method using the same information.

Structured isolation is also conditional. Sensor masking requires redundancy among the unmasked measurements and inputs. Actuator classification is an input-response compatibility statement. A process deviation aligned with the normal input-response subspace is indistinguishable from an actuator deviation, as stated in Theorem 5. In such cases, the correct output is nonunique isolation rather than a forced physical label.

C. Smoothness and Domain Validity

The Jacobian analysis requires differentiability on the operating domain used in the bound. Smooth activations satisfy this requirement directly. For piecewise-linear networks that cross activation boundaries, interval Jacobian bounds, piecewise-affine analysis, or conservative generalized-Jacobian estimates are required. Jacobian regularization during training does not replace online domain checks.

IX. CONCLUSION

This paper developed protected attention–Koopman–T–S fuzzy references for normal-data-only fault detection and structured isolation. The long nominal branch is initialized from a delay-confirmed normal anchor and is subsequently propagated without measurement correction, so new measurements cannot update its lifted state, attention context,

premise variables, or fuzzy firing strengths during warning and alarm intervals. A start-time-indexed residual unifies free multistep training and online monitoring, and its exact nonlinear propagation was established without a fault model. The complete fuzzy Koopman Jacobian exposed the additional finite-horizon amplification caused by rule overlap, local-model dispersion, and premise sensitivity. This analysis generated a recursive normal-reference radius and, together with separately calibrated stochastic remainders, a residual-dynamics-based online threshold that explicitly depends on the realized input, protected fuzzy operating region, and reference age. Full-dimensional residual statistics avoid unobservable directions created by fault-prior-based projection, while sensor masking and normal input-response decomposition provide deliberately limited structured isolation. The resulting framework preserves the stated normal-data-only information boundary and makes the fuzzy structure operational in reference generation, finite-horizon analysis, threshold construction, and isolation.

APPENDIX A

PROOF OF THE FUZZY JACOBIAN BOUND

For unit vectors $\mathbf{a}$ and $\mathbf{b}$, let $\alpha_i = \mathbf{a}^\top(\mathbf{v}_i - \bar{\mathbf{v}})$ and $\beta_i = (\mathbf{g}_i - \bar{\mathbf{g}})^\top \mathbf{J}_{\rho,z} \mathbf{b}$. Then

$$|\mathbf{a}^\top \mathbf{C}_\rho \mathbf{J}_{\rho,z} \mathbf{b}| = \left| \sum_i h_i \alpha_i \beta_i \right| \quad (65)$$

$$\leq \left(\sum_i h_i \alpha_i^2 \right)^{1/2} \left(\sum_i h_i \beta_i^2 \right)^{1/2} \quad (66)$$

$$\leq \sqrt{\lambda_{\max}(\mathbf{V})} \sqrt{\lambda_{\max}(\mathbf{J}_{\rho,z}^\top \mathbf{G} \mathbf{J}_{\rho,z})}. \quad (67)$$

Taking the supremum over $\mathbf{a}$ and $\mathbf{b}$ proves (35).

APPENDIX B

CONDITIONAL-INNOVATION FORM OF THE JOINT STATISTIC

A block Cholesky factorization of the conditional covariance gives an equivalent sequential whitening of $\mathbf{r}^y$, $\mathbf{r}^d$, and $\mathbf{r}^L$. Consequently,

$$T_k^{\text{inst}} = \|\mathbf{w}_k^y\|_2^2 + \|\mathbf{w}_k^{d|y}\|_2^2 + \|\mathbf{w}_k^{L|y,d}\|_2^2. \quad (68)$$

This decomposition is algebraically equivalent to the full Mahalanobis statistic and avoids double counting normal cross-correlation while retaining an interpretable order of residual propagation.

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